

Experiment progress recap

Symbolic-Aware Large Audio Language Models

A controlled path toward compact, interpretable symbolic streams for UniAudio-style audio-language modeling.

Source record: `doc/summary/260612_exp_journey_key_records.md`

Research Objective

Build a **learned symbolic planning token**, s_{plan} , that compresses MIDI-derived structure into a low-rate stream and can eventually condition a Large Audio-Language Model.

Current conservative claim

s_{plan} is a compact symbolic codec with acoustic-token utility and MIR-recoverable structure under controlled Slakh experiments.

Notation boundary: s_{plan} denotes the current learned symbolic planning token. s_{LLM} denotes the planned LLM-supervised symbolic tokenizer/token stream, and has not been trained yet.

Not yet claimed: s_{LLM} training, robust final LALM reasoning, or controllable generation.

UniAudio2.0 Background

Background: All experiments in this deck are designed around the UniAudio2.0 formulation: a unified autoregressive audio-language model that uses ReasoningCodec to factorize audio into reasoning tokens and reconstruction/semantic acoustic tokens.

Why it matters here: Our S_{plan} stream is not a generic MIDI add-on. It is designed as a symbolic planning stream that can be compared against UniAudio2.0-style reasoning and semantic acoustic-token streams under aligned, shuffled, and dummy controls.

Design consequence: We keep the released UniAudio2.0 stack as the anchor and ask whether a compact learned symbolic stream provides additional music-structural information beyond the frozen audio-token baseline.

UniAudio2.0 References

Citation: Yang, Wang, Chong, Liu, Wu, and Meng. “UniAudio 2.0: A Unified Audio Language Model with Text-Aligned Factorized Audio Tokenization.” arXiv:2602.04683, 2026.

Resource	Link
Paper	arxiv.org/abs/2602.04683
Demo	dongchaoyang.top/UniAudio2Demo
Code	github.com/yangdongchao/UniAudio2
Checkpoints	huggingface.co/Dongchao/UniAudio2_ckpt

Presentation rule: cite UniAudio2.0 at the beginning, then separate our claim: S_plan is the current symbolic-planning stream; S_{LLM} is planned and not trained yet.

Interactive Web Demo Companion

Purpose: The Overleaf/Beamer deck cannot reliably play audio examples, so qualitative examples are organized as a separate web demo in the same style as task-oriented audio model demos.

Demo link: <https://dabinkim0.github.io/symbolic-fused-lalm-experiments/demo/>

Demo sections: (1) UniAudio2.0 Audio / Speech Tasks, (2) UniAudio2.0 Music Tasks, and (3) Symbolic-Aware MIDI / Music Tasks.

Claim boundary: The demo uses playable Slakh clips and current artifacts for S_plan. It marks final integrated LALM preservation, decoded generation, and S_{LLM} examples as missing until those experiments are actually run.

Demo Update: Evidence Dashboard Structure

Update objective: The demo page was reorganized from a metric dump into an evidence dashboard that separates held-out aggregate results, validation case studies, and future generation slots.

Demo block	What was added	Claim boundary
UniAudio2.0 Audio / Speech	Released UniAudio2 baseline slots for TTS, text-to-sound, and instructed TTS	Preservation check only; no Symbolic-Aware generation checkpoint yet
UniAudio2.0 Music	EXP025 token utility and EXP029 stream-control evidence now separated from downstream claims	Mechanistic proxy, not decoded music improvement
Symbolic-Aware MIDI / Music	EXP024 feature-reconstruction gallery and EXP031A MIR task framing	Symbolic feature/MIR evidence, not MIDI transcription or audio generation

Experiment Ladder I: Diagnostics To S_plan

Stage		Role in the contribution	Status
Frozen	UniAudio	Establish baseline limits and avoid overclaiming QA	Completed diagnostic
Token-level	MIR	Check whether released audio tokens contain recoverable music structure	Completed diagnostic
EXP020–023		Audit symbolic representations and length/control constraints	Completed groundwork
EXP024		Learn compact S _p lan RVQ codec	Completed
EXP025		Test acoustic-token utility before LALM integration	Positive proxy

Reading order: these stages justify why the project moves from frozen UniAudio diagnostics to a compact learned symbolic token instead of raw MIDI appending.

Experiment Ladder II: Interface And Supervision

Stage	Role in the contribution	Status
EXP027/028	Diagnose failed or weak LALM interfaces	Completed diagnostics
EXP029	Test UniAudio-style symbolic stream interface	Small positive control result
EXP030	Build MIR/text target bank for tokenizer supervision	Completed data layer
EXP031A	MIR-supervised S _p lan structure recovery	Baseline complete; controls incomplete
EXP031A controls	Test zero-feature and label-shuffle alternatives before causal claims	Required next Planned, not run
EXP031B	LLM-supervised S _{LLM} tokenizer training	

Claim order: EXP031A controls precede EXP031B because they decide whether current S_plan structure recovery is causal or mostly prior-driven.

Evidence Gaps To Keep Visible

- ▶ **EXP031A controls incomplete:** zero-feature and label-shuffle controls are required before making a causal MIR-structure claim.
- ▶ **No S_LLM tokenizer yet:** EXP031A is MIR-supervised S_plan evaluation, not Llama/Qwen-supervised token training.
- ▶ **Instrument macro-F1 is nonmonotonic:** it peaks around 350–400 steps and drops afterward, while other MIR heads keep improving.
- ▶ **Final LALM usage not proven:** EXP029 is a stream-interface proxy, not final open-ended music reasoning.
- ▶ **Decoded music downstream metrics incomplete:** music captioning, generated-audio quality/control, and paper-compatible QA/GPT-score style evaluations remain future work.
- ▶ **No generation intervention yet:** editing S_plan and observing controlled audio changes remains future work.

Dataset Protocol

Split	Tracks	30 s windows	Notes
Train	1,289	11,279	Track-level split before windowing
Validation	270	2,334	Same conversion protocol
Test	151	1,385	Held-out tracks, zero overlap

- ▶ Slakh supports controlled paired audio–MIDI experiments with objective symbolic labels.
- ▶ All main symbolic comparisons preserve the same split and paired-data budget.
- ▶ **Real-domain symbolic-token generalization is still future work.**

Metrics 01A: Current Evidence Axes

Question: Yes, we have evaluated music-related downstream behavior, but not every metric is final paper-grade downstream evidence. Current evidence is strongest where labels and controls are objective.

Metric axis	What it answers	Current status
Output-level music QA	Can frozen UniAudio2 answer music questions from audio?	Completed diagnostic on Slakh instrument presence
Token-level causal proxy	Does aligned S_plan reduce C_a uncertainty?	Completed for EXP025/EXP029-style CE/PPL
MIR-structured recovery	Are instruments, pitch-class, meter, key, density, and polyphony recoverable from S_plan?	EXP031A baseline complete; controls incomplete

Metric rule: use objective metrics and negative controls first; treat broad open-ended downstream tasks as future evidence until the output protocol is stable.

Metrics 01B: Future Evidence Axes

Purpose: This slide separates planned decoded-output evidence from the completed diagnostic/proxy metrics so the deck does not overstate the current claim.

Metric axis		What it answers	Current status
Decoded generation/control		Does symbolic conditioning improve generated audio and controllability?	Not yet evaluated
Caption/open metrics	QA	Does final LALM improve music captioning, MMAU-like QA, or GPT-scored outputs?	Not yet stable enough for claims

Presentation rule: future decoded examples should be shown after EXP031A controls, not before, because the control suite determines whether the symbolic stream itself carries causal structure.

Metrics 02: Output-Level Music QA

What was evaluated: Frozen UniAudio2 was tested on Slakh primary instrument-presence QA. This is a music-related downstream output task, but it is used as a diagnostic baseline rather than the main contribution result.

Example prompt type: “You are given a music audio clip. Answer whether the target instrument is present. Return only yes or no.”

Metrics used: Exact accuracy, macro-F1, active invalid rate after conservative semantic recovery, strict invalid rate before semantic recovery, and prediction distribution.

Observed result: 151 clips / 906 QA rows; exact accuracy 0.6181; macro-F1 0.3782; active invalid 0.1181; strict invalid 0.3400; 796 yes / 3 no / 107 invalid.

Interpretation: this exposed prompt-following and yes-bias issues, so QA alone is not a safe proof of music understanding or symbolic-token benefit.

Missing evaluation: structural closed-set QA beyond primary instrument presence, MusicNet-style real-recording QA, caption attribute F1, and constrained decoding stability must still be run before broad output-level claims.

Metrics 03: Token-Level Causal Metrics

Purpose: Measure whether the symbolic stream changes token prediction in a controlled way, before claiming decoded downstream behavior.

Stage	Metric	Control design	Interpretation
EXP025	Semantic CE/PPL for C_a	reason-only, low-rate, aligned S_plan, shuffled S_plan, dummy	Aligned S_plan gives best test CE 7.5963 and PPL 1990.88
EXP027	Test CE/PPL	aligned vs shuffled vs dummy row insertion	Negative result: aligned/shuffled/dummy are indistinguishable
EXP028	CE/PPL/accuracy and adapter gate	LoRA adapter vs tiny-gate warm start	Stable but near-zero gate; not symbolic-use evidence
EXP029	Stream-native CE/accuracy	aligned vs shuffled vs dummy symbolic stream	Small controlled gain: aligned test CE 5.7663 vs shuffled 5.7847

Metrics 04: MIR-Structured Symbolic Metrics

Purpose: Measure whether `S_plan` preserves objective music structure from Slakh MIDI-derived labels, rather than only reconstructing internal features or lowering acoustic CE.

Metric families: Instrument macro/micro F1, pitch-class macro-F1, meter accuracy, weak key accuracy, segment density accuracy, segment polyphony accuracy, and numeric MIR losses.

EXP031A result: At test step 625: loss 6.6153, instrument macro-F1 0.5014, instrument micro-F1 0.7908, pitch-class macro-F1 0.8541, meter 0.7834, weak key 0.5567, segment density 0.8743, segment polyphony 0.7614.

Metric caveat: Instrument macro-F1 is better at step 375 than 625, so rare-class thresholding/calibration must be handled separately from broad MIR recovery.

Interpretation: EXP031A is the strongest current music-structure metric block, but it is MIR-supervised `S_plan` evaluation, not LLM-supervised `S_LLM` training.

Missing evaluation: zero-feature and label-shuffle controls, per-class threshold calibration, and real-domain transfer are required before claiming robust symbolic MIR understanding.

Metrics 05A: Missing Downstream Plan – Understanding

Downstream task	Metrics to report	Required experiment
Music captioning / symbolic-to-text	caption attribute F1, tag mAP, CIDEr/BLEU/ROUGE/BERTScore, CLaMP3 retrieval	Run after EXP030 text targets are connected to a decoder
Audio-to-symbolic	symbolic CE/PPL, token accuracy, instrument F1, pitch-range MAE, density/polyphony error	Train audio-to-S_plan or MIR target head with controls
Paper-compatible broad QA	exact/F1/invalid, MMAU-style accuracy, optional GPT-score with fixed rubric	Run after output protocol and parser stability improve

Priority: these tasks connect the symbolic stream to human-readable music reasoning, but only after the current EXP031A control boundary is clean.

Metrics 05B: Missing Downstream Plan – Generation

Downstream task	Metrics to report	Required experiment
Symbolic-to-audio decoded output	FAD/KL, CLAP or CLaMP3 consistency, instrument classifier F1, chroma/beat consistency	Decode selected checkpoints and compare aligned vs shuffled S_plan
Generation control intervention	control adherence, edit locality, audio quality, before/after symbolic edit metrics	Run symbolic intervention suite; no current evidence

Next metric priority: finish EXP031A controls, then add per-sample metric joins across QA, token CE/PPL, and MIR recovery before broad decoded-generation claims.

Diagnostic 01: Frozen UniAudio2 QA Boundary

EXP Objective: Measure whether the released UniAudio2 checkpoint can serve as a reliable music-understanding QA baseline before adding symbolic tokens.

Results: On 151 Slakh clips and 906 instrument-presence QA rows, exact accuracy was 0.6181, macro-F1 was 0.3782, active invalid rate was 0.1181, and strict invalid rate was 0.3400. Predictions were heavily yes-biased: 796 yes, 3 no, and 107 invalid.

Discussion: The model often produced caption-like fragments rather than exact yes/no outputs, so raw QA accuracy mixes musical understanding, formatting compliance, and label-prior bias. This makes frozen QA useful as a diagnostic boundary, but too brittle as the main contribution evidence.

Key point: Frozen UniAudio2 QA is not the final claim; it motivates controlled token-level and symbolic-control evaluation.

Next Step: Move to token-level MIR probes, where objective labels and shuffled controls can test whether audio tokens contain recoverable music structure.

Diagnostic 02: Token-Level MIR Probes

EXP Objective: Audit whether UniAudio2 reason and semantic audio tokens contain recoverable musical structure even when final QA behavior is brittle.

Results: Frozen-token probe families showed that MIR information is recoverable from reason tokens, semantic tokens, or their combination under controlled probe settings. Later label-shuffle and test-shuffle controls clarified which probe results were meaningful.

Discussion: This stage changed the evaluation philosophy. Instead of relying on one open-ended QA score, the project needed information-recovery probes, negative controls, and aligned-vs-corrupted symbolic comparisons.

Key point: The audio token space is not empty; symbolic fusion can be evaluated by whether it adds structured information beyond recoverable audio-token content.

Next Step: Design symbolic inputs that fit the UniAudio token budget and can be tested against shuffled or dummy symbolic controls.

EXP020–023: Symbolic Representation Feasibility

EXP Objective: Determine whether raw MIDI-style tokens, rule summaries, or learned low-rate symbolic tokens are the right representation for UniAudio-style fusion.

Results: Raw MIDI tokenizations were often too long or brittle for clean 30 s window comparisons. Low-rate rule summaries were more controllable, averaging about 149 tokens per 30 s with 0% truncation, but they remained hand-designed and limited.

Discussion: These experiments created the first major design fork. Simply appending raw MIDI would make gains hard to interpret because length, context capacity, and alignment artifacts would confound the result. A compact learned token was therefore academically cleaner.

Key point: The project pivoted from “append MIDI” to “learn a low-rate symbolic planning token.”

Next Step: Train a learned symbolic bottleneck, `S_plan`, with fixed length and explicit codebook-health diagnostics.

EXP024: Learned S_{plan} Reconstruction Codec

EXP Objective: Train a compact symbolic planning codec that compresses MIDI-derived frame features into four low-rate RVQ streams for later UniAudio-style conditioning.

Results: The long checkpoint at step 42,000 reached validation reconstruction loss 0.00918, binary accuracy 0.99777, continuous MSE 0.00172, and codebook usage 0.3134 / 0.7187 / 0.8073 / 0.8124.

Discussion: EXP024 established that a learned symbolic bottleneck is feasible and not collapsed. However, reconstruction alone only proves that MIDI-derived structure can be compressed; it does not prove that the token helps audio modeling or language reasoning.

Key point: S_{plan} became a technically healthy symbolic codec, but not yet a reasoning token.

Next Step: Test whether aligned S_{plan} reduces uncertainty in UniAudio semantic acoustic tokens compared with shuffled and dummy controls.

EXP024 Demo Evidence: Feature-Level Gallery

Demo update: The web demo now shows three fixed validation windows as target / reconstruction / absolute-error heatmaps, rather than using a piano-roll proxy that visually overlapped with MIR examples.

Fixed example	Binary acc.	Cont. MSE	Unique RVQ codes per book
Slakh_Track01501_w000_s10000000	0.00174	15 / 15 / 17 / 17	
Slakh_Track01501_w001_s10003000	0.00246	20 / 24 / 26 / 25	
Slakh_Track01501_w003_s10009000	0.00130	17 / 17 / 21 / 25	

Claim boundary: this is symbolic feature reconstruction evidence. It is not waveform generation, full MIDI event reconstruction, or audio-to-MIDI transcription.

EXP025: Acoustic-Token Utility Proxy

EXP Objective: Evaluate whether fixed S_{plan} carries information useful for predicting UniAudio semantic acoustic tokens C_a from reason tokens R_a .

Results: Aligned S_{plan} gave the best proxy result: valid CE 7.5073, test CE 7.5963, and test PPL 1990.88. It improved over reason-only test CE 7.6573 and shuffled S_{plan} test CE 7.6307.

Discussion: This is the first clean positive result for the learned symbolic token. Because aligned symbolic tokens beat shuffled and dummy controls, the improvement is less likely to be explained by merely adding tokens or capacity. Still, the task is a proxy objective, not final LALM reasoning.

Key point: Aligned S_{plan} reduces acoustic-token uncertainty under controls.

Next Step: Integrate S_{plan} into the LALM-facing path and test whether the final model interface can actually use the symbolic stream.

EXP025 Demo Evidence: Aggregate First, Preview Second

Demo update: The page now explicitly separates the held-out aggregate result from validation fixed-preview token tables. The aggregate CE/PPL row carries the claim; fixed examples are sanity previews.

Condition	Valid CE	Test CE	Test PPL	Role
Reason only	7.5650	7.6573	2116.06	Audio-token-only baseline
Low-rate rule summary	7.5534	7.6445	2089.16	Rule-symbolic bridge
Aligned S _{plan}	7.5073	7.5963	1990.88	Best proxy result
Shuffled S _{plan}	7.5376	7.6307	2060.58	Alignment-corrupted control
Dummy/random controls	7.54–7.69	7.63–7.69	2064–2196	Extra-token/context controls

Do not call this downstream generation improvement. It is token-level utility evidence for why the symbolic stream should help.

EXP027: Naive LALM Row Insertion

EXP Objective: Test the simplest integration path: insert or append learned symbolic rows into the released UniAudio2 LALM interface with minimal architectural changes.

Results: Reason-only achieved test CE 5.3598 and PPL 212.67. Aligned `S_plan` degraded to test CE 5.5113, while shuffled `S_plan` was 5.5119 and dummy symbolic tokens were 5.5100.

Discussion: Aligned, shuffled, and dummy symbolic inputs were nearly indistinguishable, which means the LALM did not use the symbolic content in this interface. This negative result is important: it separates a good symbolic codec from a usable LALM integration.

Key point: Naive symbolic insertion failed; the bottleneck shifted from token learning to interface design.

Next Step: Try a gentler adapter path that can inject symbolic information without destabilizing the released LALM representation space.

EXP028: Adapter Interface Diagnostic

EXP Objective: Diagnose whether LoRA plus a symbolic adapter can create a non-destructive route for symbolic information into the LALM-facing prediction objective.

Results: The first LoRA+adapter setup destabilized training: valid CE 7.8022, valid PPL 2445.86, and accuracy 0.0284. A tiny-gate warm start recovered reason-only-level behavior with valid CE 5.4116 and accuracy 0.1391.

Discussion: The tiny-gate result shows that the interface can be made non-destructive, but the adapter gate stayed around 3.36×10^{-4} and the delta ratio around 10^{-6} . Therefore this experiment does not prove symbolic usage.

Key point: Stability is possible, but the adapter is effectively too closed to count as symbolic fusion evidence.

Next Step: Redesign the symbolic stream to follow UniAudio-style multi-stream structure rather than treating symbols as an external side-channel.

EXP029: UniAudio-Style Symbolic Stream

EXP Objective: Create a stream-native symbolic interface using 8 audio streams, 4 symbolic RVQ streams, and 1 text stream, ordered as $R_a \rightarrow S_plan \rightarrow C_a$.

Results: The sequence layout was healthy: 151 reason frames, 30 symbolic frames, 376 semantic frames, total length 557, max length 1024, 0% truncation, and 0% bad-position rate. Aligned S_plan achieved test CE 5.7663, beating shuffled 5.7847 and dummy 5.7852.

Discussion: The improvement is modest, but it is the cleanest LALM-interface evidence so far because alignment matters and extra symbolic rows alone are insufficient. The result supports stream-native fusion, while still falling short of final open-ended LALM reasoning.

Key point: A UniAudio-compatible symbolic stream produces a small but controlled aligned-vs-corrupted gain.

Next Step: Build structured MIR/text targets so the symbolic tokenizer can be supervised more directly and interpreted beyond acoustic CE.

EXP029 Demo Evidence: Case Spectrum

Demo update: The stream-control card now shows both positive and boundary fixed examples. This prevents the qualitative section from looking cherry-picked and makes the small-margin nature of EXP029 transparent.

Validation fixed example	Aligned loss	Best control	Margin	Interpretation
Slakh_Track01501_w000_500000000	5.280		+0.054	Strong positive fixed-case margin
Slakh_Track01501_w001_500000000	6.163		+0.004	Small positive fixed-case margin
Slakh_Track01501_w002_500000000	6.091		-0.006	Boundary / failure case
Slakh_Track01501_w003_500000000	5.869		-0.007	Boundary / failure case

Claim boundary: EXP029 supports a stream-interface mechanism. It is not broad decoded-music generation and not final text-to-music behavior.

EXP030: MIR/Text Target Bank

EXP Objective: Construct the supervision layer needed to train and evaluate `S_plan` as a structured symbolic token rather than only a reconstruction or acoustic-prediction aid.

Results: The target bank produced 89,630 train LLM examples and 67,072 train QA examples from 11,279 windows, plus 18,517 / 13,849 valid examples and 11,012 / 8,242 test examples. There were 0 conversion failures across splits.

Discussion: EXP030 is not a model-result claim by itself. Its contribution is data infrastructure: it links symbolic/audio windows to MIR attributes and text-like supervision targets, enabling MIR-supervised `S_plan` evaluation now and planned LLM-supervised `S_LLM` training later.

Key point: EXP030 turns the project from codec evaluation into supervised symbolic-token learning.

Next Step: Run MIR-supervised `S_plan` evaluation first, because it gives objective, controllable labels before expensive LLM-supervised `S_LLM` training.

EXP031A: Objective And Configuration

EXP Objective: Test whether structured MIR attributes can be recovered from `S_plan` when the EXP024 autoencoder is initialized and effectively frozen. This is MIR-supervised `S_plan` evaluation, not LLM-supervised `S_LLM` training.

Configuration: EXP024 `S_plan` encoder initialized and treated as the symbolic representation source; MIR heads evaluate instruments, pitch classes, key, meter, density, and polyphony.

Claim boundary: This stage evaluates structured recovery from the current symbolic planning token. It does not train a language-readable `S_LLM` tokenizer and does not prove final LALM generation.

Key point: MIR structure is recoverable from `S_plan`, but control runs are required before the paper can claim causal symbolic content.

EXP031A: Claim-Validation Results

EXP Objective: Evaluate whether MIR recovery reflects sample-specific S_{plan} content rather than dataset priors, head bias, or class imbalance.

Comparison	Split step	/	Inst. macro	Inst. micro	mi-Meter	Weak key	Interpretation
Aligned S_{plan}	test / 625		0.5014	0.7908	0.7834	0.5567	Main positive row; insufficient alone for causal claim.
Zero features	valid 500	/	0.4263	0.8354	0.8548	0.1264	Reveals strong prior/head effects for micro-F1 and meter.
Label shuffle	needed	-	-	-	-	-	Required to test feature-target alignment.
Target prior	planned	-	-	-	-	-	Required floor for common instruments, meter, and density.

Discussion: The main aligned row is promising, but the zero-feature control shows that several high MIR scores can be explained without symbolic input. EXP031A should therefore be reported

EXP031A Controls: Axis-Level Claim Boundary

EXP Objective: Decide which MIR axes are safe contribution evidence and which axes are prior-dominated diagnostics.

Metric axis	Aligned valid 625	Zero-feature valid 500	Current interpretation
Weak key	0.5484	0.1264	Most clearly feature-dependent so far; still needs label-shuffle and test confirmation.
Instrument macro-F1	0.4986	0.4263	Some symbolic signal, but class imbalance and thresholding remain unresolved.
Pitch-class macro-F1	0.8514	0.8295	Small positive gap; descriptive until target-prior baseline is reported.
Instrument micro-F1	0.8047	0.8354	Do not headline; zero-feature exceeds aligned.
Meter accuracy	0.7661	0.8548	Do not headline; common-meter prior is strong.
Segment density	0.8873	0.8794	Nearly matched by zero-feature; use only with controls and examples.

EXP031A Controls: Execution Order

Next Step 1: Run zero-feature first with the same B4096/S625 evaluation boundary. If it performs strongly, revisit MIR head priors before claiming symbolic content.

Next Step 2: Run label-shuffle next. If shuffled labels approach aligned performance, the feature-target alignment claim weakens.

Next Step 3: Run threshold calibration on validation outputs before changing the instrument loss. If calibration recovers macro-F1, balanced BCE may be unnecessary.

Priority rule: these controls are more urgent than balanced-inst training or S_LLM decoder ablations because they protect the core academic claim.

EXP031B: Planned S_{LLM} Tokenizer

EXP Objective: Train a planned LLM-supervised symbolic tokenizer, S_{LLM} , using an LLM decoder objective so that symbolic tokens become language-readable and can support text/MIR supervision beyond hand-structured heads.

Results: **Not run yet.** The planned first decoder family is Llama-3.2-compatible, starting with 300M for stable/debug runs and checking 3B later. Newer Llama or Qwen comparisons should be treated as ablations after the control suite is clean.

Key point: S_{LLM} is the next conceptual tokenizer stage, but not the next safest experiment until controls pass.

EXP031B: Ordering Rationale

Discussion: Running S_LLM before EXP031A controls would create an interpretability problem: a high-capacity decoder might learn dataset priors or target shortcuts, making it unclear whether the current S_plan representation itself carries causal structure.

Decoder ablation plan: After controls pass, compare Llama-3.2-300M, Llama-3.2-3B, newer Llama, and Qwen as decoder/supervisor ablations. The first comparison should test stability and claim clarity, not only instruction-following strength.

Next Step: After EXP031A controls validate aligned S_plan content, run EXP031B S_LLM with Llama-3.2-300M first, then compare stronger decoders as an academic ablation.

EXP034B: Music-Claim Hardening – Complete

EXP Objective: Open-gate matched comparison: reason-only versus aligned, shuffled, and dummy S_plan streams under identical LoRA / data / step budget (800 steps, seed 260635). All four conditions complete.

Condition	Test CE	Test PPL	Test Acc	Δ CE vs aligned
Reason-only baseline	5.8590	350.4	0.1039	+0.259
Aligned S _p lan stream	5.5998	270.4	0.1207	—
↪ Shuffled S _p lan	5.6144	274.3	0.1191	+0.015
↪ Dummy stream	5.6143	274.3	0.1193	+0.015

Claim (paper-safe): Aligned S_plan beats both alignment-corrupted controls by ≈ 0.015 CE and the no-symbolic reason-only baseline by 0.259 CE. Gate value is higher for aligned (0.177) than shuffled (0.167) or dummy (0.163), consistent with active symbolic use. `claim_ready = true`.

Claim boundary: EXP034B is semantic-token stream training under oracle symbolic input. It is not text-to-music, song-generation, or MIDI-to-audio generation.

EXP035: Decoded Rollout Bridge

EXP Objective: Convert EXP029/EXP034 stream checkpoints into the first decoded generation artifacts by rolling out semantic acoustic tokens and decoding them to waveform.

Generation contract: $R_a + S_plan + \text{semantic BOS} \rightarrow \text{autoregressive } C_a \text{ rollout} \rightarrow \text{codec decode}$.

Condition	Role	Status
Reason-only	Audio-token generation baseline	Available after source checkpoint completes
Aligned S_plan	Oracle symbolic-plan-conditioned generation	Next export target
Shuffled S_plan	Corrupted alignment control	Run after checkpoint completes
Dummy stream	Token-budget control	Needed before strong claim

Important: EXP035 is not end-to-end text-to-music. It is the fastest decoded bridge proving that the symbolic stream can reach waveform outputs.

Final LALM Training Readiness

Current status: The file `train_symbolic_fused_lalm.py` exists, but it currently validates data/batch contracts and writes a resolved run spec. It does not yet run full UniAudio2 symbolic-fused LoRA/FSDP optimization.

Stage	Claim target	Readiness
G0	$R_a + S_{plan} \rightarrow C_a$ rollout + waveform	Mostly ready through EXP035 export
G1	MIDI / $S_{plan} \rightarrow$ audio generation	Requires reducing or removing R_a dependence
G2	Text-to-music / song generation	Requires prompt supervision and final generation trainer wiring

Implementation next step: wire the real trainer backend: UniAudio2 checkpoint load, LoRA injection, symbolic stream/adaptor injection, autoregressive audio-token loss, checkpoint save/reload and EXP035-compatible state dicts.

Current Paper Story

1. Frozen UniAudio2 is not a reliable final music-QA baseline, but its token streams carry recoverable structure.
2. Raw symbolic MIDI is too long or brittle for a clean UniAudio-style fusion interface.
3. S_plan gives a compact learned symbolic stream with healthy reconstruction and codebook diagnostics.
4. Aligned S_plan improves acoustic-token prediction compared with shuffled/dummy controls (EXP025).
5. Naive LALM insertion fails; a stream-native warmup (EXP029) gives a small positive control result.
6. MIR-supervised EXP031A suggests S_plan contains structured musical attributes, with axis-level caveats.
7. **EXP034B (complete):** In the hardened open-gate comparison with all four matched conditions, aligned S_plan achieves test CE 5.600 vs. shuffled 5.614, dummy 5.614, and reason-only 5.859. This is the strongest controlled positive result in the ladder.

Most Important Next Experiments

Rank	Experiment	Why it matters	Command status
P0	EXP034B controls	Makes aligned vs shuffled vs dummy stream comparison paper-safe	Complete: aligned 5.600 < shuffled 5.614 < dummy 5.614 < reason-only 5.859
P0	EXP035 decoded rollout bridge	Turns token-stream checkpoints into waveform examples for the demo	Run after EXP034B checkpoints finish
P1	EXP031A zero/label-shuffle controls	Separates MIR priors/head bias from symbolic content	Complete before stronger MIR claims
P1	Final trainer backend wiring	Required for true Symbolic-Aware generation LALM training	Not wired yet; dry-run contract exists
P2	EXP031B S_LLM supervision	Tests language-readable symbolic-token training	Start after controls pass

Team Handoff Wording

What we can say

We have a compact learned symbolic planning codec, `S_plan`, that (1) reconstructs MIDI-derived structure faithfully (EXP024), (2) improves acoustic-token prediction under aligned-vs-corrupted controls in three independent experiments (EXP025, EXP029, EXP034B), and (3) supports MIR structure recovery on pitch-class, density, and key axes pending label-shuffle controls (EXP031A). EXP034B is the strongest result: all four matched conditions complete, aligned beats both controls by ≈ 0.015 CE.

What we should not say yet

`S_LLM` has already been trained, or `S_plan` is final-LALM-proven / generation-controllable.

End State For The Next Paper Iteration

Strongest current evidence

- ▶ EXP024 reconstruction and codebook health.
- ▶ EXP025 aligned improvement over corrupted controls.
- ▶ EXP029 stream-native positive gap.
- ▶ **EXP034B complete:** aligned $5.600 <$ both controls $5.614 <$ reason-only 5.859 ; gate opens for aligned.
- ▶ EXP031A MIR recovery baseline (pitch-class, density, key).

Missing before stronger claims

- ▶ EXP031A zero-feature and label-shuffle controls.
- ▶ Calibration for instrument macro-F1.
- ▶ EXP031B S_LLM LLM-supervised tokenizer training.
- ▶ Final LALM and generation interventions.